

PRINCIPAL ENTERPRISE ARCHITECT REFERENCE SERIES

VOLUME 4 OF 8

Artifact Catalog & Quality Attributes

Every artifact an Architecture Review Board should expect to see — purpose, owner, consumers, lifecycle, and automation — paired with a modern quality attribute taxonomy that goes beyond generic NFRs to cover AI explainability, agent reliability, and memory consistency.

Enterprise Architecture Review Board Handbook · Banking & Financial Services Edition

Part A — The Complete Artifact Catalog

An ARB that doesn't have a clear, enforced artifact catalog ends up with wildly inconsistent submission quality — some teams over-document trivial decisions, others under-document significant ones, and reviewers waste cycles asking for missing information rather than evaluating substance. This catalog defines what "good" looks like for every artifact a mature banking ARB should expect across the full lifecycle, from business case through retirement.

HOW TO READ THIS CATALOG

Each artifact entry covers five dimensions: **Purpose** (what decision or understanding it enables), **Owner** (who is accountable for its accuracy), **Consumers** (who relies on it), **Lifecycle** (when it's created, updated, retired), and **Automation** (whether and how it can be generated or validated automatically rather than hand-maintained).

7.1 Strategy & Vision Artifacts

Artifact	Purpose	Owner	Consumers	Automation
Business Case	Justifies investment with cost, benefit, and risk quantification	Business sponsor	Executive Steering Committee, Finance	Low — fundamentally a judgment artifact, though cost models can pull from FinOps tooling
Vision Document	Articulates the desired future state and why it matters strategically	Business sponsor + Chief Architect (joint)	ARB, delivery teams, broader organization	None — narrative artifact
Architecture Vision	Translates business vision into target architecture direction and guiding principles	Chief Architect / Lead Solution Architect	ARB, all contributing architects	Low — references the pattern catalog and reference architectures from Volume 3
Capability Map	Traces the initiative to business capabilities affected (see Volume 3, Part B)	Business capability owner + Enterprise Architect	ARB, capability redundancy analysis	Medium — generated/updated from the knowledge graph

7.2 Architecture Description Artifacts

Artifact	Purpose	Owner	Consumers	Automation
Context Diagram	Shows the system's boundary and external interactions (C4 Level 1 or equivalent)	Solution Architect	ARB, integration partner teams	Medium — can be partially generated from API gateway/service mesh metadata
Container Diagram	Shows the major deployable units and their interactions (C4 Level 2)	Solution Architect	ARB, delivery team, operations	Medium — generatable from infrastructure-as-code definitions
Deployment Diagram	Maps the architecture onto physical/cloud infrastructure topology	Cloud/Platform Architect	ARB, CCoE, operations, security	High — should be generated directly from IaC (Terraform/CloudFormation) rather than hand-drawn, to guarantee accuracy
Threat Model	Identifies attack surfaces, threat actors, and mitigations (e.g., STRIDE-based analysis)	Security Architect	Cyber Security Council, ARB	Medium — threat modeling tools can pre-populate from architecture diagrams, but expert review remains essential

7.3 Decision & Risk Artifacts

Artifact	Purpose	Owner	Consumers	Automation
ADR (Architecture Decision Record)	Captures a single significant decision, its context, options considered, and rationale	Decision-making architect	ARB, future architects (via knowledge graph), auditors	Medium — AI-assisted drafting from decision context is increasingly viable (see Volume 7) graph), auditors
Risk Register	Tracks identified architectural and technology risks with likelihood, impact, and mitigation status	Risk owner (often the Solution Architect, escalating to Risk function for material risks)	ARB, Risk Committee, Internal Audit	Medium — risk scoring can be templated; identification requires human judgment
Decision Log	Chronological record of all ARB decisions, distinct from individual ADRs — the meeting-level audit trail	ARB Secretariat / Chair	Auditors, Risk Committee, new ARB members onboarding	High — should be a direct, automatically timestamped system output, not manually transcribed
Compliance Matrix	Maps architecture decisions/controls to specific regulatory requirements (see Volume 6 for banking-specific regulatory mapping)	Compliance Architect / Risk & Compliance liaison	Regulators, Internal Audit, Risk Committee	Medium — control-to-regulation mapping can be templated and reused across initiatives in the same regulatory domain

7.4 Integration & Interface Artifacts

Artifact	Purpose	Owner	Consumers	Automation
Integration Contract	Defines the agreed interface, data, and behavioral contract between two systems/teams	Both integrating teams (jointly owned)	ARB, both delivery teams, testing/QA	High — contract testing tooling can validate compliance automatically and continuously post-approval
API Specification	Formal, typically OpenAPI/Swagger-based definition of a REST API's structure and behavior	API-owning team	API consumers, API gateway, developer portal	High — should be the literal source of truth the gateway and documentation are generated from, not a parallel hand-maintained document
AsyncAPI	Equivalent specification standard for event-driven/asynchronous interfaces	Event-publishing team	Event consumers, event catalog/registry	High — same principle as API Specification, applied to event schemas

7.5 AI-Specific Artifacts

Artifact	Purpose	Owner	Consumers	Automation
Prompt Library	Versioned, tested collection of system prompts and prompt templates used in production AI features	AI Solution Architect / Prompt Engineer	AI Governance Board, Responsible AI Council, delivery teams reusing patterns	High — should be under version control with the same rigor as code, with automated regression testing against known good/bad outputs
Memory Policy	Defines what an AI agent or system remembers across interactions, retention duration, and access boundaries	AI Solution Architect	Data Governance Council, AI Governance Board, Privacy Office	Low — policy definition is a judgment artifact, though enforcement can be automated
Agent Specification	Defines an AI agent's scope of autonomy, available tools/actions, escalation triggers, and human-override mechanisms	AI Solution Architect	ARB, AI Governance Board, Responsible AI Council, operations	Medium — see Volume 7 for the full Agent Specification template
MCP Tool Contract	Defines the interface, permissions, and behavioral contract for a tool exposed to AI agents via Model Context Protocol or equivalent	Tool-owning team	Agent developers, security (tool permissions are a security-critical surface), ARB	High — should be machine-validated against actual tool implementation, given the security sensitivity of agent tool access
A2A Contract	Defines the interaction protocol and trust boundary between two autonomous AI agents (agent-to-agent)	Both agent-owning teams (jointly)	ARB, AI Governance Board, security	Medium — an emerging artifact type as multi-agent architectures mature; tooling is still developing as of 2026

7.6 Operational & Lifecycle Artifacts

Artifact	Purpose	Owner	Consumers	Automation
Runbooks	Step-by-step operational procedures for known scenarios (incident response, failover, recovery)	Operations/SRE team, with architect input on architecture-specific procedures	On-call operations staff, incident commanders	Medium — common scenarios can be automated as runbook-as-code (e.g., automated failover scripts), reducing reliance on manual procedure execution under pressure
Support Model	Defines who supports the system, escalation tiers, SLAs, and handoff between development and operations	Service Owner	Operations, ARB (to confirm sustainability before approval), CAB	Low — organizational/process artifact
Retirement Checklist	Ensures clean, complete decommissioning — data archival/deletion per retention policy, dependency verification, access revocation	Service Owner, with Data Governance and Security sign-off	ARB, Data Governance Council, Internal Audit	Medium — dependency verification can be automated via the knowledge graph (Volume 3); data handling steps require careful manual verification given regulatory consequences of getting this wrong

THE MOST COMMONLY SKIPPED ARTIFACT

The Retirement Checklist is, by a wide margin, the most frequently skipped artifact in this catalog — decommissioning rarely has an executive sponsor pushing for rigor the way a new build does, and the consequences of skipping it (orphaned data, undiscovered dependencies, incomplete access revocation) often surface years later as audit findings or security incidents rather than immediately. Treating Retirement Checklist completion as a mandatory ARB gate, with the same rigor as new-build approval, is one of the highest-leverage low-cost governance improvements available to a Principal Architect.

Part B — Architecture Quality Attributes

Generic non-functional requirements ("the system should be fast and secure") are useless as governance criteria because they aren't testable. This part defines twenty-one quality attributes with banking-relevant, testable framing for each — including six AI/agent-specific attributes that don't appear in pre-2023 architecture references and that most ARB charters still haven't formally incorporated.

8.1 Classical Quality Attributes

Attribute	Testable Framing	Banking-Specific Note
Performance	Response time, throughput, and resource utilization under specified load, expressed as percentiles (p50/p95/p99), not averages	Peak-period scenarios (month-end, year-end, market-open) often dominate, not steady-state averages
Availability	Uptime percentage with explicit measurement window and what counts as "down" (full outage vs. degraded)	Payment and trading systems frequently carry contractual or regulatory availability SLAs distinct from generic best-effort targets
Scalability	Defined scaling dimension (users, transactions, data volume) and the mechanism (horizontal/vertical) with concrete headroom targets	Should explicitly address both gradual growth and sudden demand spikes (e.g., a viral product launch or market event)
Security	Specific control objectives (authentication, authorization, encryption at rest/in transit, audit logging) mapped to a recognized framework (NIST, ISO 27001)	Must explicitly address regulatory-driven controls (PCI-DSS for payments, data residency for cross-border operations)
Privacy	Data minimization, purpose limitation, and subject rights (access, deletion, portability) as concrete, testable controls	GDPR and equivalent regimes impose specific, auditable obligations distinct from general security posture
Maintainability	Mean time to implement a defined class of change, code complexity metrics, test coverage thresholds	Particularly material for core banking systems with multi-decade expected lifespans
Operability	Ease of routine operation — deployment frequency achievable, monitoring/alerting completeness, runbook coverage	Should be assessed against the Support Model artifact (Section 7.6) at review time, not assumed
Deployability	Deployment frequency achievable, rollback time, blast radius of a failed deployment	Banking change-freeze windows (e.g., around regulatory reporting dates) constrain this and should be designed around explicitly
Observability	Coverage of metrics, logs, and traces sufficient to diagnose an unknown failure mode without code changes	Increasingly a regulatory expectation itself — operational resilience regimes expect demonstrable real-time visibility into critical system health
Recoverability	Recovery Time Objective (RTO) and Recovery Point Objective (RPO), tested via actual recovery drills, not just documented targets	Subject to formal regulatory testing requirements in many jurisdictions (operational resilience testing regimes)
Auditability	Completeness and tamper-evidence of the audit trail for a defined set of sensitive actions	A first-class design requirement in banking, not an afterthought — should be reviewed with the same rigor as security
Compliance	Explicit mapping to applicable regulations with control evidence (see Compliance Matrix, Section 7.3)	Should be assessed per-jurisdiction for multi-jurisdiction banks, since requirements diverge materially
Portability	Effort required to move the system to a different infrastructure provider or environment	Directly relevant to vendor-concentration risk management (see Volume 2, Section 3.6)
Usability	Task completion rate and time-on-task for defined user journeys, typically measured via usability testing	Customer-facing and internal operations/back-office usability are both relevant and frequently under-prioritized for the latter

8.2 AI & Agent-Specific Quality Attributes

These seven attributes are the newest additions to the architecture quality vocabulary, and the area where ARB charters most commonly lag behind actual practice. A Principal Architect operating in 2026 should treat fluency here as a baseline expectation, not a specialization.

Attribute	Testable Framing	Banking-Specific Note
AI Explainability	Ability to produce a human-understandable rationale for a specific AI-driven decision, at a level of detail proportionate to the decision's stakes	Credit decisioning and similar high-stakes AI use cases typically face explicit regulatory explainability requirements (adverse action notice equivalents) that constrain model architecture choices, not just downstream reporting
AI Fairness	Measured outcome parity across protected/relevant demographic groups against defined fairness metrics (demographic parity, equalized odds, etc., chosen deliberately since these metrics can conflict with each other)	Fair lending regulatory regimes impose specific obligations; the choice of fairness metric itself should be a documented, defensible decision, not a default
AI Robustness	Performance degradation under adversarial input, distribution shift, or edge cases, tested via deliberate red-teaming and stress testing	Fraud detection and similar adversarial-environment AI use cases face active, adaptive adversaries, unlike most traditional software quality attributes
Safety	Bounded worst-case behavior — what is the AI system structurally prevented from doing, regardless of input or failure mode	Should be addressed through architectural constraints (tool permission scoping, action approval gates) rather than relying solely on model behavior, which cannot be guaranteed with certainty
Trustworthiness	A composite, harder-to-test attribute combining calibrated confidence (the system's stated confidence matches its actual accuracy) with consistent behavior over time	Increasingly a named requirement in AI governance frameworks (NIST AI RMF's "trustworthy AI" characteristics) even where not yet precisely testable
Agent Reliability	Task completion rate for defined agent workflows, including graceful failure and appropriate escalation when the agent cannot complete a task, rather than silent failure or fabricated success	Particularly critical where agents take real-world actions (initiating payments, modifying records) rather than only generating text, given the higher cost of an unreliable action versus an unreliable response
Memory Consistency	An AI agent's recalled context/memory remains accurate and internally consistent across a session and across sessions where persistence is intended	A novel attribute with essentially no pre-2023 precedent; testing methodology is still maturing across the industry, and ARBs should expect to evolve their review criteria as practice matures

CONTEXT FRESHNESS — AN EIGHTH EMERGING ATTRIBUTE

Closely related to Memory Consistency but distinct: whether the information an AI system is reasoning over (retrieved documents, tool outputs, cached context) reflects current reality versus stale data. For RAG-based architectures and AI agents acting on enterprise data, a context-freshness SLA (e.g., "retrieved data must be no more than 15 minutes stale for transaction-related queries") should be a named, testable requirement — most current AI architecture reviews don't yet ask this question explicitly, and it is becoming a more frequent source of production incidents as the gap between this practice and increasing AI deployment scale widens.

8.3 Resolving Quality Attribute Trade-offs

Many of these attributes are in direct tension — Security and Usability frequently trade off against each other; Performance and Auditability can conflict when comprehensive audit logging adds latency; AI Explainability and raw model performance often trade off, since the most explainable model architectures aren't always the most accurate ones. This is precisely the territory ATAM and CBAM (Volume 2, Part B) are designed to navigate systematically rather than leaving these trade-offs to be resolved implicitly and inconsistently by whichever engineer happens to be making the call in the moment.

A GOVERNANCE DISCIPLINE WORTH ADOPTING

For any architecturally significant initiative, require an explicit statement of which quality attributes were deliberately prioritized and which were consciously deprioritized, with rationale — rather than a checklist where every attribute is rated "addressed." A design that explicitly says "we are accepting reduced AI Explainability in exchange for materially better fraud detection accuracy, because [rationale], and this trade-off has been reviewed by [Responsible AI Council]" is far more defensible under audit than one that silently doesn't mention explainability at all.